

ML
Unsupervised
1st Exam
Q&A

Contents

1 st Exam	2
Cumulants	2
KL Divergence.....	5
Marginal probabilities, entropies, and mutual information for a joint distribution of 2 variables.....	8
Parameter estimation and Maximum Likelihood.....	11
Unsupervised Machine Learning.....	12
Expectation Maximization (EM) and Maximum Entropy	13
Principal Component Analysis (PCA).....	14
Kernel PCA.....	19
Independent Component Analysis (ICA)	20
2 nd Exam	21
Factor Analysis	21
K- Means	24
t- distributed stochastic Neighbourhood Embedding. T-SNE... ..	25
Scaling and Projection Methods	25
Non-negative Matrix Factorization (NMF)	26
Mixture Models:.....	27
KL Divergence.....	30
Clustering	31
Biclustering.....	32
Multivariate Gaussian	33

1st Exam

Cumulants

Assume you have two independent random variables X and Y , which are uniformly distributed on $[-1, 1]$, i.e.

$$p_X(x) = p_Y(x) = \begin{cases} \frac{1}{2} & \text{if } x \in [-1, 1] \\ 0 & \text{else.} \end{cases}$$

As in the lecture, let $\kappa_n(X)$ denote the n -th order cumulant of a random variable X . Compute the quantities and answer the questions below.

Hint: Recall that $\kappa_4(X) = E(X^4) - 3(E(X^2))^2$ if X is centered.

You can use a pocket calculator. Round to three decimal places. Number format example: 0.667

- $\kappa_4(X) =$ ✓
- $\kappa_4(2X) =$ ✓
- $\kappa_4(X + Y) =$ ✓
- X is ☒ Gaussian

Handwritten calculations for cumulants of a uniform distribution on $[-1, 1]$:

$$\kappa_4 = E(x^4) - 3(E(x^2))^2$$
$$E(x^2) = \int_{-1}^1 x^2 p(x) dx = \int_{-1}^1 x^2 \cdot \frac{1}{2} dx = \left[\frac{x^3}{3} \cdot \frac{1}{2} \right]_{-1}^1 = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$
$$E(x^4) = \int_{-1}^1 x^4 p(x) dx = \int_{-1}^1 x^4 \cdot \frac{1}{2} dx = \left[\frac{x^5}{5} \cdot \frac{1}{2} \right]_{-1}^1 = \frac{1}{10} + \frac{1}{10} = \frac{2}{10} = \frac{1}{5}$$
$$\kappa_4(x) = \frac{1}{5} - 3 \cdot \left(\frac{1}{3}\right)^2 = \frac{9 - 3 \cdot 5}{45} = -\frac{6}{45} = -0,133$$
$$\kappa_4(2X) = 2^4 \kappa_4(x) = 16 \cdot (-0,133) = -2,133$$
$$\kappa_4(X+Y) = \kappa_4(X) + \kappa_4(Y) = -0,133 - 0,133 = -0,267$$

sub - Gauss.

3. Cumulant of $X + Y$:

$$\kappa_4(X + Y) = E((X + Y)^4) - 3(E((X + Y)^2))^2$$

Expanding $(X + Y)^4$ using the binomial theorem:

$$(X + Y)^4 = X^4 + 4X^3Y + 6X^2Y^2 + 4XY^3 + Y^4$$

Since X and Y are independent, $E(XY) = E(X)E(Y) = 0$ and $E(X^3Y) = E(X^3)E(Y) = 0$.

Thus, only the terms X^4 , $6X^2Y^2$, and Y^4 will contribute.

$$= E(X^4) + 6E(X^2)E(Y^2) + E(Y^4)$$

As X and Y are uniform over $[-1, 1]$, $E(X^2) = E(Y^2) = \frac{1}{3}$, $E(X^4) = E(Y^4) = \frac{1}{5}$.

$$= \frac{1}{5} + 6 \left(\frac{1}{3}\right) \left(\frac{1}{3}\right) + \frac{1}{5} = -0.267$$

4. Sub-Gaussian or Super-Gaussian?

Given the value of $\kappa_4(X)$ is negative, X is sub-Gaussian.

Assume you have two independent, discrete random variables X and Y , which follow independent symmetric Bernoulli distributions taking only the values $+1$ and -1 , both with probability $1/2$. As in the lecture, let $\kappa_n(X)$ denote the n -th order cumulant of a random variable X . Compute the quantities below.

Hint: Recall that $\kappa_3(X) = E(X^3)$ and $\kappa_4(X) = E(X^4) - 3(E(X^2))^2$ if X is centered.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

- $\kappa_3(X) =$

✖

- $\kappa_4(X) =$

✖

- $\kappa_4(3X)/\kappa_4(X) =$

✖

- $\kappa_4(X+Y)/\kappa_4(X) =$

✖

$$E(X^2) = (1)^2 \times \frac{1}{2} + (-1)^2 \times \frac{1}{2} = \frac{1}{2} + \frac{1}{2} = 1$$

With this correction:

1. Compute $\kappa_3(X)$:

$$\kappa_3(X) = E(X^3) = (1)^3 \times \frac{1}{2} + (-1)^3 \times \frac{1}{2} = 1 \times \frac{1}{2} - 1 \times \frac{1}{2} = 0$$

2. Compute $\kappa_4(X)$:

$$\kappa_4(X) = E(X^4) - 3(E(X^2))^2$$

$$\kappa_4(X) = 1 - 3 \times (1)^2 = 1 - 3 = -2$$

3. Compute $\kappa_4(3X)/\kappa_4(X)$:

$$\kappa_4(3X) = 3^4 \times \kappa_4(X) = 81 \times (-2) = -162$$

$$\frac{\kappa_4(3X)}{\kappa_4(X)} = \frac{-162}{-2} = 81$$

4. Compute $\kappa_4(X+Y)/\kappa_4(X)$:

$$\kappa_4(X+Y) = \kappa_4(X) + \kappa_4(Y) = (-2) + (-2) = -4$$

$$\frac{\kappa_4(X+Y)}{\kappa_4(X)} = \frac{-4}{-2} = 2$$

So, the corrected results are:

- $\kappa_3(X) = 0$
- $\kappa_4(X) = -2$
- $\frac{\kappa_4(3X)}{\kappa_4(X)} = 81$
- $\frac{\kappa_4(X+Y)}{\kappa_4(X)} = 2$

1. Compute $\kappa_3(X)$:

Since X follows a symmetric Bernoulli distribution, X takes the values +1 and -1 with equal probability. Therefore,

$$\kappa_3(X) = E(X^3) = (1)^3 \cdot \frac{1}{2} + (-1)^3 \cdot \frac{1}{2} = 1 \cdot \frac{1}{2} - 1 \cdot \frac{1}{2} = 0$$

2. Compute $\kappa_4(X)$:

Using the hint provided,

$$\kappa_4(X) = E(X^4) - 3(E(X^2))^2$$

Since X takes values +1 and -1 with equal probability, $E(X) = 0$ and $E(X^2) = (1)^2 \cdot \frac{1}{2} + (-1)^2 \cdot \frac{1}{2} = \frac{1}{2}$. Therefore,

$$\kappa_4(X) = E(X^4) - 3\left(\frac{1}{2}\right)^2$$

Now, $E(X^4) = (1)^4 \cdot \frac{1}{2} + (-1)^4 \cdot \frac{1}{2} = \frac{1}{2} + \frac{1}{2} = 1$.

$$\kappa_4(X) = 1 - 3 \times \left(\frac{1}{2}\right)^2 = 1 - 3 \times \frac{1}{4} = 1 - \frac{3}{4} = \frac{1}{4}$$

3. Compute $\kappa_4(3X)/\kappa_4(X)$:

$$\kappa_4(3X) = 3^4 \kappa_4(X) = 81 \times \frac{1}{4} = \frac{81}{4}$$

$$\text{Therefore, } \frac{\kappa_4(3X)}{\kappa_4(X)} = \frac{\frac{81}{4}}{\frac{1}{4}} = 81$$

4. Compute $\kappa_4(X + Y)/\kappa_4(X)$:

Since X and Y are independent, we can use the property that $\kappa_4(X + Y) = \kappa_4(X) + \kappa_4(Y)$.

$$\kappa_4(X + Y) = \kappa_4(X) + \kappa_4(Y) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$\text{Therefore, } \frac{\kappa_4(X+Y)}{\kappa_4(X)} = \frac{\frac{1}{2}}{\frac{1}{4}} = 2$$

So, the results are:

- $\kappa_3(X) = 0$
 - $\kappa_4(X) = 0.25$
 - $\frac{\kappa_4(3X)}{\kappa_4(X)} = 81$
 - $\frac{\kappa_4(X+Y)}{\kappa_4(X)} = 2$
-

Question 4

Incorrect

Mark 0.00 out of 9.00

You are given a dataset with n samples and three features. The covariance matrix $\mathbf{C}$ has eigenvalues $(\lambda_1, \lambda_2, \lambda_3) = (5, 3, 1)$ and associated normalized eigenvectors $(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$, $\|\mathbf{v}_i\| = 1$. Calculate

$$A = (2\mathbf{v}_1^T + \mathbf{v}_2^T + 3\mathbf{v}_3^T)\mathbf{C}(3\mathbf{v}_2 + \mathbf{v}_3)$$

A=

✗

Since the eigenvectors are orthogonal (because they correspond to different eigenvalues), their inner products will be zero:

$$\mathbf{v}_i^T \mathbf{v}_j = 0, \text{ for } i \neq j$$

With this in mind, we can calculate the expression:

$$(2\mathbf{v}_1^T + \mathbf{v}_2^T + 3\mathbf{v}_3^T) \mathbf{C} (3\mathbf{v}_2 + \mathbf{v}_3) = 2\mathbf{v}_1^T \mathbf{C} \mathbf{v}_2 + \mathbf{v}_2^T \mathbf{C} \mathbf{v}_2 + 3\mathbf{v}_3^T \mathbf{C} \mathbf{v}_2 + 2\mathbf{v}_1^T \mathbf{C} \mathbf{v}_3 + \mathbf{v}_2^T \mathbf{C} \mathbf{v}_3 + 3\mathbf{v}_3^T \mathbf{C} \mathbf{v}_3$$

Since $\mathbf{v}_i^T \mathbf{v}_j = 0$ for $i \neq j$, the terms that have mixed indices will become zero, leaving us with:

$$= 3\mathbf{v}_2^T \mathbf{C} \mathbf{v}_2 + 3\mathbf{v}_3^T \mathbf{C} \mathbf{v}_3$$

Now, use the fact that $\mathbf{v}_2$ is an eigenvector associated with eigenvalue 3, and $\mathbf{v}_3$ is an eigenvector associated with eigenvalue 1. This means:

$$\mathbf{C} \mathbf{v}_2 = 3\mathbf{v}_2, \mathbf{C} \mathbf{v}_3 = 1\mathbf{v}_3$$

Substitute these into the expression:

$$= 3\mathbf{v}_2^T * 3\mathbf{v}_2 + 3\mathbf{v}_3^T * \mathbf{v}_3 = 9\mathbf{v}_2^T \mathbf{v}_2 + 3\mathbf{v}_3^T \mathbf{v}_3$$

As we know that $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$ (generally for any vector $\mathbf{v}$), we get:

$$= 9*1 + 3*1 = 12$$

KL Divergence

Assume you have two random variables Y and Z , which are uniformly distributed on $[0, \theta_Y]$ and $[0, \theta_Z]$, respectively, i.e.

$$p_Y(x) = \begin{cases} \frac{1}{\theta_Y} & \text{if } x \in [0, \theta_Y] \\ 0 & \text{else.} \end{cases}$$

$$p_Z(x) = \begin{cases} \frac{1}{\theta_Z} & \text{if } x \in [0, \theta_Z] \\ 0 & \text{else.} \end{cases}$$

For $\theta_Y = e^5$ and $\theta_Z = e^{10}$, compute the KL divergence

$$D_{\text{KL}}(p_Y || p_Z) = \int_{-\infty}^{\infty} p_Y(x) \log[p_Y(x)/p_Z(x)] dx.$$

As for "log", use the natural logarithm.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

$$D_{\text{KL}}(p_Y || p_Z) =$$

$$\begin{aligned}
 D_{\text{KL}}(p_Y \| p_Z) &= \int_0^{\theta_Y} \frac{1}{\theta_Y} \log \left(\frac{\frac{1}{\theta_Y}}{\frac{1}{\theta_Z}} \right) dx \\
 &= \int_0^{\theta_Y} \frac{1}{\theta_Y} \log \left(\frac{\theta_Z}{\theta_Y} \right) dx \\
 &= \frac{1}{\theta_Y} \log \left(\frac{\theta_Z}{\theta_Y} \right) \int_0^{\theta_Y} dx \\
 &= \log \left(\frac{\theta_Z}{\theta_Y} \right)
 \end{aligned}$$

The KL divergence between two continuous uniform distributions $p_Y(x)$ and $p_Z(x)$ is given by:

$$D_{\text{KL}}(p_Y \| p_Z) = \int_{-\infty}^{\infty} p_Y(x) \log \left(\frac{p_Y(x)}{p_Z(x)} \right) dx$$

Given that $p_Y(x)$ and $p_Z(x)$ are uniform distributions, their probability density functions are constant over their respective intervals:

$$\begin{aligned}
 p_Y(x) &= \begin{cases} \frac{1}{\theta_Y} & \text{if } x \in [0, \theta_Y] \\ 0 & \text{else} \end{cases} \\
 p_Z(x) &= \begin{cases} \frac{1}{\theta_Z} & \text{if } x \in [0, \theta_Z] \\ 0 & \text{else} \end{cases}
 \end{aligned}$$

Plugging in the values $\theta_Y = e^5$ and $\theta_Z = e^{10}$, we get:

$$\begin{aligned}
 p_Y(x) &= \begin{cases} \frac{1}{e^5} & \text{if } x \in [0, e^5] \\ 0 & \text{else} \end{cases} \\
 p_Z(x) &= \begin{cases} \frac{1}{e^{10}} & \text{if } x \in [0, e^{10}] \\ 0 & \text{else} \end{cases}
 \end{aligned}$$

Now, let's compute the KL divergence:

$$\begin{aligned}
 D_{\text{KL}}(p_Y \| p_Z) &= \int_0^{e^5} \frac{1}{e^5} \log \left(\frac{1/e^5}{1/e^{10}} \right) dx \\
 &= \int_0^{e^5} \frac{1}{e^5} \log(e^5) dx \\
 &= \int_0^{e^5} \frac{1}{e^5} \cdot 5 dx \\
 &= \frac{5}{e^5} \cdot x \Big|_0^{e^5} \\
 &= \frac{5}{e^5} \cdot e^5 - \frac{5}{e^5} \cdot 0 \\
 &= 5
 \end{aligned}$$

So, $D_{\text{KL}}(p_Y \| p_Z) = 5$.

Given two vectors:

$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

$$\mathbf{y} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$$

Compute the outer product matrix $\mathbf{A} = \mathbf{xy}^T$.

Enter the (integer) values of the third row of $\mathbf{A}$ below:

- $A_{31} =$

✖

- $A_{32} =$

✖

- $A_{33} =$

✖

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 4 & 5 & 6 \\ 8 & 10 & 12 \\ 12 & 15 & 18 \end{pmatrix}$$

Marginal probabilities, entropies, and mutual information for a joint distribution of 2 variables

Assume you are given two binary random variables X and Y taking only the values 0 and 1 where their joint probability distribution is specified by the table below:

	$Y = 0$	$Y = 1$
$X = 0$	0.25	0.125
$X = 1$	0.25	0.375

This means, that e.g. $P(X = 0, Y = 0) = 0.25$, etc.

Compute the quantities (marginal probabilities, entropies, and mutual information) below.

Use the **natural logarithm** in your calculations.

You can use a pocket calculator. Round the final values to two decimal places. (You may want to keep three-digit accuracy if you use intermediate results.)

Number format example: 0.67

• $P(X = 1) =$

✗

• $P(Y = 1) =$

✗

• $H(X) =$

✗

• $H(Y) =$

✗

• $I(X; Y) =$

✗

$$P(X=1) = 0.25 + 0.375 = 0.625$$

$$P(Y=1) = 0.125 + 0.375 = 0.5$$

$$P(X=0) = 0.25 + 0.125 = 0.375$$

$$P(Y=0) = 0.25 + 0.25 = 0.5$$

$$\begin{aligned} H(X) &= -\sum_i P(X=i) \ln P(X=i) \\ &= -(P(X=0) \ln P(X=0) + P(X=1) \ln P(X=1)) \\ &= -(0.375 \ln 0.375 + 0.625 \ln 0.625) \\ &= 0.662 \end{aligned}$$

$$\begin{aligned} H(Y) &= -\sum_i P(Y=i) \ln P(Y=i) \\ &= -(P(Y=0) \ln P(Y=0) + P(Y=1) \ln P(Y=1)) \\ &= -(0.5 \ln 0.5 + 0.5 \ln 0.5) \\ &= 0.693 \end{aligned}$$

$$\begin{aligned} H(X, Y) &= -\sum_x \sum_y P(x, y) \ln P(x, y) \\ &= -(0.25 \ln 0.25 + 0.125 \ln 0.125 + 0.25 \ln 0.25 + 0.375 \ln 0.375) \\ &= 1.321 \end{aligned}$$

$$\begin{aligned} I(X; Y) &= H(X) + H(Y) - H(X, Y) \\ &= 0.662 + 0.693 - 1.321 = 0.034 \end{aligned}$$

Assume you are given two binary random variables X and Y taking only the values 0 and 1 where their joint probability distribution is specified by the table below:

	$Y = 0$	$Y = 1$
$X = 0$	$1/3$	$1/3$
$X = 1$	0	$1/3$

This means, that e.g. $P(X = 0, Y = 0) = 1/3$, etc.

Compute the quantities (marginal probabilities, entropies, and mutual information) below. Use the natural logarithm in your calculations.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

• $P(Y = 0) =$

• $H(X) =$

• $H(Y) =$

• $I(X; Y) =$

Handwritten calculation of mutual information $I(X; Y)$:

$$\begin{aligned}
 & \begin{array}{c|cc|c} & Y=0 & Y=1 & \\ \hline X=0 & 1/3 & 1/3 & 2/3 \\ X=1 & 0 & 1/3 & 1/3 \\ \hline & 1/3 & 2/3 & \end{array} \\
 & I(X, Y) = H(X) + H(Y) - H(X, Y) = \\
 & = -\frac{2}{3} \log \frac{2}{3} - \frac{1}{3} \log \frac{1}{3} - \\
 & \quad - \frac{1}{3} \log \frac{1}{3} - \frac{2}{3} \log \frac{2}{3} - \\
 & \quad - \left[-\frac{1}{3} \log \frac{1}{3} - \frac{1}{3} \log \frac{1}{3} - \frac{1}{3} \log \frac{1}{3} \right] = \\
 & = \underline{0.6365} + \underline{0.635} - \underline{1.0386} = \\
 & = 0,1744
 \end{aligned}$$

1. Marginal probabilities:

- $P(Y = 0) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$
- $P(Y = 1) = 0 + \frac{1}{3} = \frac{1}{3}$
- $P(X = 0) = \frac{1}{3} + 0 = \frac{1}{3}$
- $P(X = 1) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$

2. Entropy:

- $H(X) = -\sum_i P(X = i) \log_2 P(X = i)$
 - $H(X) = -(P(X = 0) \log_2 P(X = 0) + P(X = 1) \log_2 P(X = 1))$
 - $H(X) = -\left(\frac{1}{3} \log_2 \frac{1}{3} + \frac{2}{3} \log_2 \frac{2}{3}\right)$
 - $H(X) = -\left(\frac{1}{3} \times (-1.58) + \frac{2}{3} \times (-0.58)\right)$
 - $H(X) = 0.64$ (rounded to two decimal places)
- $H(Y) = -\sum_i P(Y = i) \log_2 P(Y = i)$
 - $H(Y) = -(P(Y = 0) \log_2 P(Y = 0) + P(Y = 1) \log_2 P(Y = 1))$
 - $H(Y) = -\left(\frac{2}{3} \log_2 \frac{2}{3} + \frac{1}{3} \log_2 \frac{1}{3}\right)$
 - $H(Y) = -\left(\frac{2}{3} \times (-0.58) + \frac{1}{3} \times (-1.58)\right)$
 - $H(Y) = 0.64$ (rounded to two decimal places)

3. Mutual Information:

- $I(X; Y) = \sum_{x,y} P(X = x, Y = y) \log_2 \left(\frac{P(X=x, Y=y)}{P(X=x)P(Y=y)} \right)$
 - $I(X; Y) = P(X = 0, Y = 0) \log_2 \left(\frac{P(X=0, Y=0)}{P(X=0)P(Y=0)} \right) + P(X = 0, Y = 1) \log_2 \left(\frac{P(X=0, Y=1)}{P(X=0)P(Y=1)} \right)$
 - $+ P(X = 1, Y = 0) \log_2 \left(\frac{P(X=1, Y=0)}{P(X=1)P(Y=0)} \right) + P(X = 1, Y = 1) \log_2 \left(\frac{P(X=1, Y=1)}{P(X=1)P(Y=1)} \right)$
 - $I(X; Y) = \frac{1}{3} \log_2 \left(\frac{\frac{1}{3}}{\frac{1}{3} \times \frac{2}{3}} \right) + \frac{1}{3} \log_2 \left(\frac{\frac{1}{3}}{\frac{1}{3} \times \frac{1}{3}} \right)$
 - $+ 0 \times \log_2 \left(\frac{0}{\frac{2}{3} \times \frac{1}{3}} \right) + \frac{1}{3} \log_2 \left(\frac{\frac{1}{3}}{\frac{2}{3} \times \frac{1}{3}} \right)$
 - $I(X; Y) = \frac{1}{3} \times 0.58 + \frac{1}{3} \times 1.58 + 0 + \frac{1}{3} \times 0.58$
 - $I(X; Y) = 0.17$ (rounded to two decimal places)

So, the calculated quantities are:

- $P(Y = 0) = 0.67$
- $H(X) = 0.64$
- $H(Y) = 0.64$
- $I(X; Y) = 0.17$

$$a) \quad P(X=0) = \frac{1}{3} + \frac{1}{3} = \frac{2}{3} \quad P(Y=0) = \frac{1}{3}$$

$$P(X=1) = \frac{1}{3} \quad P(Y=1) = \frac{2}{3}$$

$$b) \quad H(X) = - \left(\frac{2}{3} \ln\left(\frac{2}{3}\right) + \frac{1}{3} \ln\left(\frac{1}{3}\right) \right) = 0,636$$

$$H(Y) = - \left(\frac{1}{3} \ln\left(\frac{1}{3}\right) + \frac{2}{3} \ln\left(\frac{2}{3}\right) \right) = 0,636$$

$$c) \quad I(X, Y) = H(X) + H(Y) - H(X, Y)$$

$$H(X, Y) = - \sum p(x, y) \ln(p(x, y)) = - \left(\frac{1}{3} \ln \frac{1}{3} + \frac{1}{3} \ln \frac{1}{3} + \frac{1}{3} \ln \frac{1}{3} + 0 \right)$$

$$= - \ln \frac{1}{3} = -(-\ln 3) = \ln 3 = 1,098$$

$$I(X, Y) = 2 \cdot 0,636 - 1,098 = 0,174$$

Parameter estimation and Maximum Likelihood

Which statements about Parameter Estimation and Maximum Likelihood are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. If the data are distributed i.i.d., the likelihood can be written as a product of individual probabilities over the data samples. [1.5]
- ☐ b. The likelihood and the log-likelihood are maximized for the same parameter value, and at that parameter value $\hat{\mathbf{w}}$, their values coincide: $\mathcal{L}(\hat{\mathbf{w}}) = \ln \mathcal{L}(\hat{\mathbf{w}})$.
- ☐ c. Parameter estimation and the maximum likelihood approach rest on the assumption that the model class is not known and estimated from the data.
- ☐ d. The maximum likelihood estimator is asymptotically unbiased and efficient, but not asymptotically consistent.
- ☒ e. The mean squared error can be written as the sum of the variance of the estimator and the squared bias of the estimator. [1.5]
- ☒ f. An estimator $\hat{\mathbf{w}}$ is unbiased if its expected value (over the training set distribution) equals the true parameter: $E(\hat{\mathbf{w}}) = \mathbf{w}$. [1.5]
- ☒ g. The likelihood $\mathcal{L}(\{\mathbf{x}\}; \mathbf{w})$ is the probability that the model with parameter $\mathbf{w}$ produces the data $\{\mathbf{x}\}$. [1.5]
- ☐ h. The mean squared error is defined as follows: $\text{mse}(\hat{\mathbf{w}}) = (\hat{\mathbf{w}} - \mathbf{w})^T (\hat{\mathbf{w}} - \mathbf{w})$.

Frage 2

Richtig

Erreichte
Punkte 50,00
von 50,00Frage
markieren

Which statements about estimation theory are true?

Multiple answers are possible!

- ☒ a. The Cramér-Rao lower bound is a lower bound for the variance of an unbiased estimator. ✓
- ☐ b. Under certain reasonable regularity conditions, the Fisher information is the expected value of the derivative of the log-likelihood.
- ☒ c. An estimator is called unbiased, if, in expectation over the data distribution, it will yield the true parameter. ✓
- ☐ d. The Fisher information quantifies how much information the model parameter w carries about the variance of the parametrized data distribution.
- ☐ e. Denoting with $\hat{w}$ the estimator of the true parameter w , the mean squared error can be written as the sum of variance and bias: $\text{mse}(w, \hat{w}) = \text{Var}(w) + \text{Bias}(\hat{w}, w)$.
- ☐ f. The Cramér-Rao lower bound equals the Fisher information (matrix).
- ☒ g. Under certain reasonable regularity conditions, the Fisher information is the negative curvature of the log-likelihood. ✗
- ☒ h. An unbiased estimator is called efficient if its variance equals the Cramér-Rao lower bound. ✓

Unsupervised Machine Learning

Which statements about Unsupervised Machine Learning are true?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Underfitting means that a model is adapted to special training characteristics such as noisy measurements or outliers.
- ☐ b. Projection methods project data into a higher-dimensional space while keeping their neighborhoods.
- ☒ c. The model selection procedure is called "training" or "learning".
- ☒ d. Examples of unsupervised methods are: PCA, ICA, Factor Analysis, k-means clustering, and mixture models.
- ☒ e. In parametric models, every parameter (vector) realization represents a model.
- ☒ f. k-nearest neighbors is a nonparametric model.
- ☐ g. Generative models do not aim at modeling the data creation process.
- ☒ h. Unsupervised Machine Learning algorithms learn patterns from unlabeled data.

Which statements about Unsupervised Machine Learning are true?

- Model selection is called "training" or "learning".
- Generative models should be able to produce the distribution of the observed data.
- In parametric models, every parameter (vector) represents a model. Factor analysis is a parametric model.
- Examples of unsupervised methods are: PCA, ICA, Factor Analysis, k-means clustering, and mixture models.
- ...

Expectation Maximization (EM) and Maximum Entropy

Which statements about Expectation Maximization (EM) and Maximum Entropy are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. The EM algorithm can be applied to hidden Markov models and mixture models.
- ☒ b. In the M-step, the model parameters are updated.
- ☐ c. The E-step increases the upper bound on the log-likelihood.
- ☒ d. In EM, one deals with problems that have hidden variables as well as model parameters.
- ☐ e. The entropy of a probability distribution is always larger equal 0 and smaller equal 1.
- ☐ f. Hidden state models in general have analytical solutions for the likelihood.
- ☒ g. In the E-step, the hidden variables parameters are updated.
- ☒ h. The Maximum Entropy approach requires minimal prior assumptions to assign a-priori-probabilities.

The correct answers are: In EM, one deals with problems that have hidden variables as well as model parameters., The EM algorithm can be applied to hidden Markov models and mixture models., The E-step increases the lower bound on the log-likelihood., Maximum Entropy requires minimal prior assumptions to assign a-priori-probabilities.

The correct answers are:

E-step tightens the lower bound of the log-likelihood,

M-step maximizes the lower bound and increases the log-likelihood,

M-step: $\{ w_{k+1} = \operatorname{argmax}_{\{w\}} F(Q,w) \}$

,

E-step: $\{ Q_{k+1} = \operatorname{argmax}_{\{Q\}} F(Q,w) \}$

Principal Component Analysis (PCA)

Erreichte Punkte 5,00 von 5,00

Which statements about Principal Component Analysis (PCA) are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. The first principal component corresponds to the largest eigenvalue in the eigendecomposition of the sample covariance matrix. 
- ☐ b. The singular values from the singular value decomposition of the data matrix $\mathbf{X}$ are equal to the eigenvalues of the eigendecomposition of $\mathbf{X}^T \mathbf{X}$.
- ☐ c. The PCA solution is unique, i.e., there is only one solution.
- ☒ d. Eigenvectors can be computed iteratively with Oja's rule. 
- ☒ e. The sample variance of the k -th projection is equal to the k -th eigenvalue of the sample covariance matrix. 
- ☒ f. The matrix of eigenvectors $\mathbf{U}$ is an orthogonal matrix. 
- ☒ g. The sample covariance matrix is symmetric and positive-semidefinite. 
- ☐ h. PCA transforms data into a new coordinate system such that the data has the largest mean value along the first coordinate (first principal component), the second largest mean along the second, and so on.

The correct answers are: Common applications of PCA include data compression, data visualization and data pre-processing. , The projected data is given by $\mathbf{Y} = \mathbf{X}\mathbf{U}$, where $\mathbf{X}$ is the (centered) original data and $\mathbf{U}$ is the matrix of eigenvectors. , One can use the singular value decomposition of the data matrix $\mathbf{X}$ to perform PCA. , The PCA solution is unique up to signs, given that eigenvalues correspond to unique eigenvectors.

Which statements about Principal Component Analysis (PCA) are true?

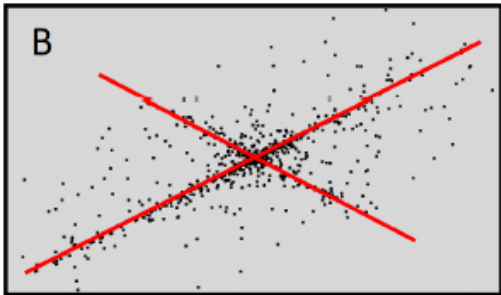
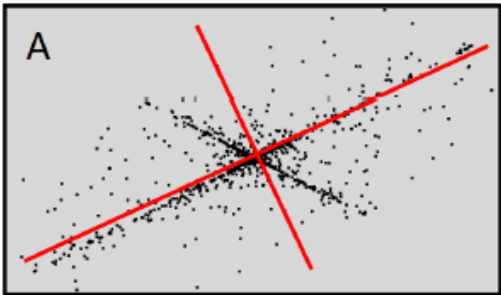
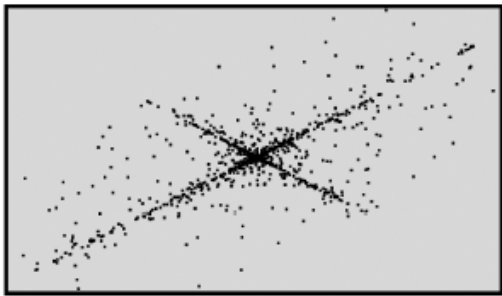
We keep the notation of the exercise slides: $\mathbf{X}$ is the data matrix. $\mathbf{U}$ is the matrix of eigenvectors of $\mathbf{X}^T \mathbf{X}$.

Multiple answers are possible!

- ☒ PCA transforms data into a new coordinate system such that the data has the largest variance along the first coordinate (first principal component), the second largest variance along the second coordinate, and so on.
- ☒ The first principal component corresponds to the largest eigenvalue in the eigendecomposition of the sample covariance matrix.
- ☐ If there are n data samples, each of which is d -dimensional, the matrix $\mathbf{U}$ has dimension $n \times d$.
- ☐ The sample covariance matrix is orthogonal and positive-semidefinite.
- ☒ The singular values from the singular value decomposition of the data matrix $\mathbf{X}$ are equal to the square root of the eigenvalues of the eigendecomposition of $\mathbf{X}^T \mathbf{X}$.
- ☐ The eigenvalues obtained from the eigendecomposition of $\mathbf{X}^T \mathbf{X}$ are all positive and sum up to 1.
- ☒ PCA is commonly used to visualize data by downprojecting it to a smaller number of dimensions, keeping the most relevant information.
- ☐ The matrix $\mathbf{U}$ is a symmetric matrix.

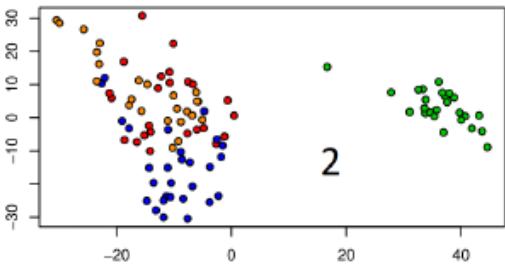
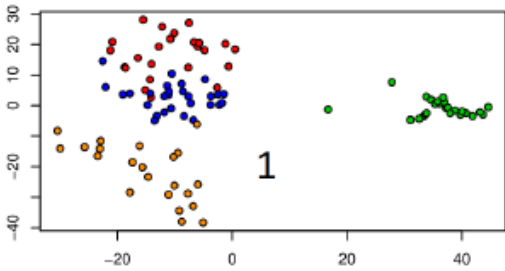
Tick the correct statements related to the pictures below:

- ICA and PCA are applied to the dataset shown in the picture in the first row (only black dots), leading to the outcomes A and B. Then picture A is the result of ☒



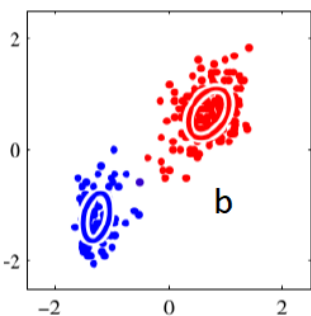
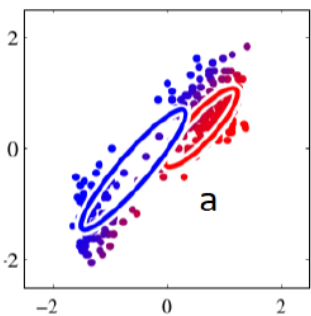
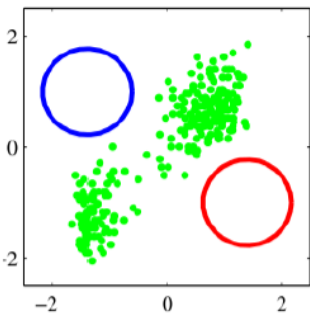
- Pictures 1 and 2 below show the result of an application of PCA to a certain dataset with 4 features in a new coordinate system given by certain principal components (PCs). The 4 different features are represented by the different colors of the dots. The PCs are ranked in the usual way and numbered accordingly. Tick the statement, which is more likely:

☒



- The picture on the left shows the original dataset where the other pictures a and b show the results of an application of the EM-algorithm with a mixture of two Gaussians to the dataset. Tick the statement, which is more likely:

☒



PCA:

You are given 4 data samples, each of dimension 3. The data matrix reads:

$$\mathbf{X} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \\ 4 & 4 & 4 \end{pmatrix}$$

The eigenvectors of $\mathbf{X}^T \mathbf{X}$ are:

$$\mathbf{u}_1 = \frac{1}{\sqrt{3}} (1, 1, 1)^T$$

$$\mathbf{u}_2 = \frac{1}{\sqrt{2}} (-1, 0, 1)^T$$

$$\mathbf{u}_3 = \frac{1}{\sqrt{2}} (-1, 1, 0)^T$$

These eigenvectors are already normalized and they are already correctly ordered corresponding to eigenvalues in descending order. (Just ignore that the second and third eigenvector are not orthogonal. This is due to the fact that the data matrix only has rank 1.)

Compute the PCA projection (matrix) $\mathbf{Y}$.

Enter the components of the projected second feature vector $\mathbf{y}_2$ into the fields below.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

- $y_2^{(1)} =$ ✖
- $y_2^{(2)} =$ ✖
- $y_2^{(3)} =$ ✖

- $y_2^{(1)} =$ ✔
- $y_2^{(2)} =$ ✔
- $y_2^{(3)} =$ ✔

To compute the PCA projection matrix $\mathbf{Y}$, you need to project the original data matrix $\mathbf{X}$ onto the eigenvectors $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$. The PCA projection matrix $\mathbf{Y}$ is given by:

$$\mathbf{Y} = \mathbf{X}\mathbf{U}$$

where $\mathbf{U}$ is the matrix formed by the eigenvectors. Since the eigenvectors are already normalized and correctly ordered, $\mathbf{U}$ is simply:

$$\mathbf{U} = [\mathbf{u}_1 \ \mathbf{u}_2 \ \mathbf{u}_3]$$

Given the data matrix $\mathbf{X}$ and the eigenvectors:

$$\mathbf{X} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \\ 4 & 4 & 4 \end{pmatrix}$$

$$\mathbf{u}_1 = \frac{1}{\sqrt{3}}(1, 1, 1)^T$$

$$\mathbf{u}_2 = \frac{1}{\sqrt{2}}(-1, 0, 1)^T$$

$$\mathbf{u}_3 = \frac{1}{\sqrt{2}}(-1, 1, 0)^T$$

We form the matrix $\mathbf{U}$:

$$\mathbf{U} = \begin{pmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix}$$

Now, to compute the PCA projection matrix $\mathbf{Y}$:

$$\mathbf{Y} = \mathbf{X}\mathbf{U}$$

Let's calculate $\mathbf{Y}$:

$$\mathbf{Y} = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \\ 4 & 4 & 4 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix}$$

Now, compute $\mathbf{Y} = \mathbf{X}\mathbf{U}$.

$$\begin{aligned} \mathbf{Y} &= \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \\ 4 & 4 & 4 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 \cdot \frac{1}{\sqrt{3}} + 1 \cdot \frac{1}{\sqrt{3}} + 1 \cdot \frac{1}{\sqrt{3}} & 1 \cdot (-\frac{1}{\sqrt{2}}) + 1 \cdot 0 + 1 \cdot \frac{1}{\sqrt{2}} & 1 \cdot (-\frac{1}{\sqrt{2}}) + 1 \cdot \frac{1}{\sqrt{2}} + 1 \cdot 0 \\ 2 \cdot \frac{1}{\sqrt{3}} + 2 \cdot \frac{1}{\sqrt{3}} + 2 \cdot \frac{1}{\sqrt{3}} & 2 \cdot (-\frac{1}{\sqrt{2}}) + 2 \cdot 0 + 2 \cdot \frac{1}{\sqrt{2}} & 2 \cdot (-\frac{1}{\sqrt{2}}) + 2 \cdot \frac{1}{\sqrt{2}} + 2 \cdot 0 \\ 3 \cdot \frac{1}{\sqrt{3}} + 3 \cdot \frac{1}{\sqrt{3}} + 3 \cdot \frac{1}{\sqrt{3}} & 3 \cdot (-\frac{1}{\sqrt{2}}) + 3 \cdot 0 + 3 \cdot \frac{1}{\sqrt{2}} & 3 \cdot (-\frac{1}{\sqrt{2}}) + 3 \cdot \frac{1}{\sqrt{2}} + 3 \cdot 0 \\ 4 \cdot \frac{1}{\sqrt{3}} + 4 \cdot \frac{1}{\sqrt{3}} + 4 \cdot \frac{1}{\sqrt{3}} & 4 \cdot (-\frac{1}{\sqrt{2}}) + 4 \cdot 0 + 4 \cdot \frac{1}{\sqrt{2}} & 4 \cdot (-\frac{1}{\sqrt{2}}) + 4 \cdot \frac{1}{\sqrt{2}} + 4 \cdot 0 \end{pmatrix} \\ &= \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 2\sqrt{3} & 0 & 0 \\ 3\sqrt{3} & 0 & 0 \\ 4\sqrt{3} & 0 & 0 \end{pmatrix} \end{aligned}$$

So the PCA projection matrix $\mathbf{Y}$ is:

$$\mathbf{Y} = \begin{pmatrix} \sqrt{3} & 0 & 0 \\ 2\sqrt{3} & 0 & 0 \\ 3\sqrt{3} & 0 & 0 \\ 4\sqrt{3} & 0 & 0 \end{pmatrix}$$

PCA:

You are given 4 data samples, each of dimension 3. The data matrix reads:

$$\mathbf{X} = \begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -3 & 0 & 3 \\ -4 & 0 & 4 \end{pmatrix}$$

The eigenvectors of the covariance matrix of this data are:

$$\mathbf{u}_1 = \frac{1}{\sqrt{2}}(-1, 0, 1)^T$$

$$\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, 0, 1)^T$$

$$\mathbf{u}_3 = (0, 1, 0)^T$$

These eigenvectors are already normalized and they are already correctly ordered corresponding to eigenvalues in descending order.

Compute the PCA projection (matrix) $\mathbf{Y}$.

Enter the components of the projected third sample $\mathbf{y}_3$ into the fields below.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

• $y_3^{(1)} =$

✖

• $y_3^{(2)} =$

✖

• $y_3^{(3)} =$

✖

$3\sqrt{2}, 0, 0$

$$\begin{pmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -3 & 0 & 3 \\ -4 & 0 & 4 \end{pmatrix} \cdot \begin{pmatrix} \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{pmatrix} = \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 2\sqrt{2} & 0 & 0 \\ 3\sqrt{2} & 0 & 0 \\ 4\sqrt{2} & 0 & 0 \end{pmatrix}$$

Kernel PCA

Tick the correct statements about Kernel PCA.

We stick to the notation from the lecture, i.e. the data points are denoted by $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathcal{X} = \mathbb{R}^m$, the kernel by $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, and the feature map by $\Phi : \mathcal{X} \rightarrow \mathcal{H}$. The feature covariance matrix is defined by $\mathbf{C} = \frac{1}{n} \sum_{i=1}^n \Phi(\mathbf{x}_i) \Phi(\mathbf{x}_i)^T$ and the Gram matrix $\mathbf{K}$ by $K_{ij} = \Phi(\mathbf{x}_i)^T \Phi(\mathbf{x}_j)$ for $i, j = 1, \dots, n$.

Wählen Sie eine oder mehrere Antworten:

- ☐ a. A disadvantage of Kernel PCA is that Φ always needs to be computed explicitly, which is computationally costly.
- ☐ b. In general, any nonnegative and symmetric function $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ defines a kernel.
- ☒ c. To compute the projection of a data point $\mathbf{x}$ onto the eigenvectors of $\mathbf{K}$, you need to center $\mathbf{x}$ and $\mathbf{K}$, as well as to normalize the eigenvectors of $\mathbf{K}$. R
- ☒ d. The main task of kernel PCA is to solve the eigenvalue problem of $\mathbf{C}$ in feature space. This can be achieved by solving the eigenvalue problem of $\mathbf{K}$ in a Euclidean space. R
- ☒ e. In general, Φ is a nonlinear map that maps into a higher dimensional feature space $\mathcal{H}$. R
- ☒ f. Linear PCA can be seen as a special case of kernel PCA by using $k(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T \mathbf{y}$ as the associated kernel function. R
- ☐ g. For kernel PCA, it is essential to center $\mathbf{K}$ before computing its eigendecomposition, since the eigenvalues of $\mathbf{K}$ can be in general be negative, which leads to inconclusive results.
- ☐ h. $\mathbf{K}$ and $\mathbf{C}$ always have the same dimensionality and the same rank.

The correct answers are: KernelPCA is an extension of PCA to nonlinear projections., Performing PCA in the feature space is equivalent to solving the eigenvalue problem for the Gram matrix.,

The Gram matrix is a square matrix which contains the kernel applied to all pairs of data points., KernelPCA maps data to a higher dimensional space using kernel methods and performs PCA on the mapped data.

Mark the correct statements about kernel PCA below (multiple might be correct).

- ☐ The kernel function $k(\cdot, \cdot)$ corresponds to a cross product in feature space.
- ☐ Kernel PCA is a non-linear classification method.
- ☒ Due to the kernel trick the feature map $\phi(\cdot)$ need never be computed explicitly.
- ☐ Kernel PCA considers an eigenvalue problem in feature space.

Last one is also correct

Independent Component Analysis (ICA)

Which statements about Independent Component Analysis (ICA) are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. Maximizing the negentropy maximizes the mutual information.
- ☒ b. In general, ICA components are neither ranked nor orthogonal. 59
- ☐ c. Non-Gaussianity can be measured by the second cumulant.
- ☒ d. Criteria for independence are mutual information between components and non-Gaussianity of components. 86
- ☐ e. The ICA solution is unique.
- ☐ f. For ICA, one needs at least as many sources as observations.
- ☐ g. Infomax maximizes the mutual information between components.
- ☒ h. Fast ICA relies on whitening and rotating the data. 8

The correct answers are: ICA is an unsupervised method of finding directions in feature space that are most statistically independent from each other., ICA requires at least as many observations as there are latent sources., FastICA is an iterative algorithm that uses whitening and rotation., INFOMAX is an algorithm that measures independence by maximizing the entropy $H(\mathbf{g}(\mathbf{y}))$, where $\mathbf{g}(\mathbf{y})$ is some elementwise function of the latent sources., Measures of independence are the mutual information between observations or the non-Gaussianity of the components., ICA is scale-invariant and its components are not ranked.

Mark the correct statements about independent component analysis (ICA) below (multiple might be correct).

- ☐ The components of ICA are ranked according to their variance.
 - ☐ Independent components are always orthogonal.
 - ☒ FastICA looks for non-Gaussianity in the data, which is measured via the kurtosis.
 - ☒ For ICA we need at least as many signals as components.
-

2nd Exam

Factor Analysis

In factor analysis part of the the covariance function is formed by

$$\mathbf{C_U} = \mathbf{U}\mathbf{U}^T = \sum_{i=1}^l \mathbf{u}_i \mathbf{u}_i^T,$$

where $\mathbf{U} \in \mathbb{R}^{m \times l}$ and $\mathbf{u}_i$ is the i -th column of $\mathbf{U}$ and $l < m$

Which constraints does this impose on the $\mathbf{C_U}$?

- ☒ a. The maximum rank of the matrix is l . ✓
- ☒ b. The number of zero eigenvalues of $\mathbf{C_U}$ is greater or equal to $m - l$. ✓
- ☒ c. Any covariance matrix in m dimensions can be written in the form above. ✗
- ☐ d. $\mathbf{C_U}$ is singular.

A, B, D is correct

Mark the correct statements about factor analysis (FA) below (multiple might be correct).

- ☒ a. FA can be viewed as a decomposition of the covariance matrix of the data. ✓
- ☐ b. The model parameters of FA are the factor loadings $\mathbf{U}$ and the factors $\mathbf{y}$ themselves.
- ☐ c. FA is usually solved using gradient descent on the likelihood function.
- ☒ d. FA is usually solved using the expectation maximization algorithm. ✓

A,D correct

Factor Analysis: Which statements are correct?

We keep the notation from the lecture:

n is the number data samples

m is the dimension of each data sample

l is the number of factors

$\mathbf{y}$ is the factor vector

$\mathbf{U}$ is the factor loading matrix

$\mathbf{\Psi}$ is the noise covariance matrix

$\mathbf{e}$ is the noise vector

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. $\mathbf{\Psi}$ has shape $m \times m$.
- ☐ b. $\mathbf{\Psi}$ has shape $l \times l$.
- ☒ c. $\mathbf{\Psi}$ has shape $n \times n$. ✗
- ☒ d. $\mathbf{U}$ has shape $m \times l$. ✓
- ☐ e. $\mathbf{U}$ has shape $m \times n$.
- ☐ f. $\mathbf{U}$ has shape $l \times m$.
- ☐ g. $\mathbf{e}$ has dimension m .
- ☐ h. $\mathbf{e}$ has dimension l .
- ☒ i. $\mathbf{e}$ has dimension n . ✗
- ☐ j. $\mathbf{y}$ has dimension m .
- ☒ k. $\mathbf{y}$ has dimension l . ✓
- ☐ l. $\mathbf{y}$ has dimension n .

A, D, G, K correct

Observations ----- $\mathbf{x} \in \mathbb{R}^m$
 Noise ----- $\boldsymbol{\epsilon} \in \mathbb{R}^m$
 Factors ----- $\mathbf{y} \in \mathbb{R}^l$
 Factor loading matrix ----- $\mathbf{U} \in \mathbb{R}^{m \times l}$
 Diagonal noise covariance matrix ----- $\boldsymbol{\Psi} \in \mathbb{R}^{m \times m}$

} parameters

Factor Analysis:

You are given the factor loading matrix $\mathbf{U}$ and the noise covariance matrix $\boldsymbol{\Psi}$:

$$\mathbf{U} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$$

$$\boldsymbol{\Psi} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

Compute the first row of the covariance matrix $\mathbf{C}$ and enter its (integer) values below.

- $C_{11} =$ ✖
- $C_{12} =$ ✖
- $C_{13} =$ ✖

Covariance Matrix $\mathbf{C}$

The covariance matrix $\mathbf{C}$ in factor analysis is given by:

$$\mathbf{C} = \mathbf{U}\mathbf{U}^T + \boldsymbol{\Psi}$$

where $\mathbf{U}^T$ is the transpose of the matrix $\mathbf{U}$.

Steps to Compute $\mathbf{C}$

1. Compute the product $\mathbf{U}\mathbf{U}^T$:

$$\mathbf{U}^T = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$$

Now, compute $\mathbf{U}\mathbf{U}^T$:

$$\mathbf{U}\mathbf{U}^T = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 2 \cdot 2 & 1 \cdot 3 + 2 \cdot 4 & 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 1 + 4 \cdot 2 & 3 \cdot 3 + 4 \cdot 4 & 3 \cdot 5 + 4 \cdot 6 \\ 5 \cdot 1 + 6 \cdot 2 & 5 \cdot 3 + 6 \cdot 4 & 5 \cdot 5 + 6 \cdot 6 \end{pmatrix} = \begin{pmatrix} 5 & 11 & 17 \\ 11 & 25 & 39 \\ 17 & 39 & 61 \end{pmatrix}$$

2. Add the noise covariance matrix Ψ :

$$C = UU^T + \Psi = \begin{pmatrix} 5 & 11 & 17 \\ 11 & 25 & 39 \\ 17 & 39 & 61 \end{pmatrix} + \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 10 & 11 & 17 \\ 11 & 30 & 39 \\ 17 & 39 & 66 \end{pmatrix}$$

3. Extract the first row of C :

$$\text{First row of } C = [10, 11, 17]$$

Mark the correct statements about factor analysis (FA) below (multiple might be correct).

- ☐ The model parameters of FA are the factor loadings U and the factors y themselves.
- ☒ FA can be viewed as a decomposition of the covariance matrix of the data.
- ☒ FA is usually solved using the expectation maximization algorithm.
- ☐ FA is usually solved using gradient descent on the likelihood function.

B,C correct

- model parameters are typically the factor loadings U and the noise variances Ψ . The factors y are latent variables and not considered model parameters.
- While gradient descent can be used in various optimization problems, it is not the typical approach for solving factor analysis problems. The EM algorithm is more commonly used because it handles the latent variables more effectively.

Which statements about Factor Analysis (FA) are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. The factors are unique only up to orthogonal transformations (rotations).
- ☒ b. Correlations between observations can only be explained by factors since the noise covariance matrix Ψ is diagonal. ✓
- ☒ c. FA describes the variability of observations in terms of latent variables and noise. ✓
- ☐ d. The factor loading matrix U describes the independent part (of the covariance of the observations).
- ☒ e. FA decomposes the covariance matrix of the data as $C = UU^T + \Psi$ ✓
- ☐ f. FA assumes Gaussian noise and non-Gaussian latent variables.
- ☐ g. Factors explain the variance of the observations, while the correlation is explained by noise.

A,B,C,E correct

K- Means

Assume you want to do k -means on the following 2-dimensional dataset:

$$X = \{(1, 0), (0, 1), (-1, 0), (0, -1), (2, 0), (2, 3), (5, 0)\}.$$

The initial cluster centers are $\mu = (0, 1)$ and $\nu = (2, 2)$.

Compute the coordinates of the new cluster centers $\mu^{\text{new}} = (\mu_1^{\text{new}}, \mu_2^{\text{new}})$ and $\nu^{\text{new}} = (\nu_1^{\text{new}}, \nu_2^{\text{new}})$ after one iteration of k -means.

Hint: Make a sketch first. The problem can mainly be solved graphically.

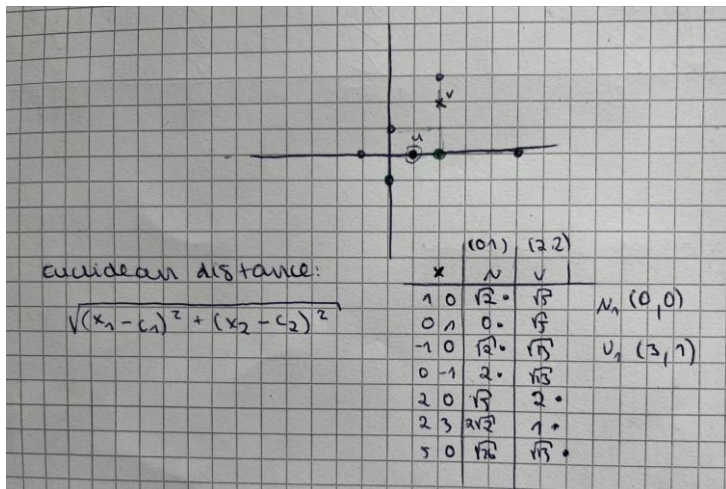
$\mu_1^{\text{new}} =$

✓, $\mu_2^{\text{new}} =$

✓
 $\nu_1^{\text{new}} =$

✓, $\nu_2^{\text{new}} =$

✓



1. Write the X in a column
2. Compute Euclidean distance for both u and v for every point
3. Assign each data point to nearest cluster center
4. To find new u and v take the average of x and y in X.

Which statements about k-means and affinity propagation are correct?

Multiple answers are possible!

- ☒ The k-means objective is to minimize the within-cluster sum of squared deviations.
- ☐ In affinity propagation: The larger the preference the smaller is the number of cluster centers.
- ☐ Affinity propagation: The larger the damping factor, the faster the convergence.
- ☒ In affinity propagation, cluster centers are data points.
- ☒ Affinity propagation: The similarity between two points is typically chosen as their negative squared distance.
- ☐ k-means always converges to the global optimum.
- ☐ The parameter k in k-means is automatically chosen by the algorithm to best fulfill the optimization objective.
- ☒ k-means is sensitive to initialization of the cluster centers.

A, D, E, H correct

- In affinity propagation, the preference parameter influences the number of clusters. A higher preference value tends to increase the number of clusters, while a lower preference value tends to decrease the number of clusters.
- The damping factor in affinity propagation helps stabilize the algorithm and can slow down convergence to avoid oscillations. A larger damping factor usually means slower convergence but more stability.
- k-means converges to a local optimum, and its solution can depend on the initial choice of cluster centers. Different initializations can lead to different results.
- The number of clusters, k , must be specified by the user and is not automatically chosen by the k-means algorithm.

t- distributed stochastic Neighbourhood Embedding. T-SNE...

Which statements about t-distributed Stochastic Neighborhood Embedding are correct? t-SNE...

- ☒ a. ...models the neighborhood in both observation and embedding space by a Gaussian distribution. ✖
- ☒ b. ...minimizes the Kullback-Leibler divergence between distributions in observation space and embedding space. ✔
- ☒ c. ...models the neighborhood of a data point in the observation space proportional to a Student's t-distribution. ✖
- ☒ d. ...is usually trained by gradient descent. ✔

B, D are correct

Scaling and Projection Methods

Which statements about Scaling and Projection Methods are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. Isomap uses a geodesic distance matrix, whose eigenvectors are the coordinates of the projected space. ✔
- ☐ b. The objective in t-SNE is minimization of the KL divergence between the similarities of the (high-dimensional) inputs and the (low-dimensional) mapped outputs. ✔
- ☒ c. In locally linear embedding (LLE), each data point is described as a linear combination of its neighbors. ✔
- ☐ d. The objective of scaling is to project data into a higher-dimensional space, e.g. to allow better model selection.
- ☐ e. In self-organizing maps the objective is to minimize an energy (error) function.
- ☐ f. Multidimensional scaling (MDS) keeps the distances between data points (as well as possible).
- ☐ g. LLE in general performs linear mappings to realize neighborhood-preserving embeddings.
- ☒ h. In t-SNE, the similarity between data points i and j is given by their joint probability. ✖

A,B,C,F, are correct

- The objective of scaling methods like MDS or PCA is to reduce the dimensionality of data while preserving important structures, not to project data into a higher-dimensional space.
- In t-SNE, the similarity between data points in the high-dimensional space is modeled by joint probabilities, but the low-dimensional embedding is found by minimizing the KL divergence between these high-dimensional joint probabilities and those in the low-dimensional space.

Which statements about projection and scaling methods are correct?

- ☒ a. Non-negative Matrix Factorization (NMF) performs a multiplicative decomposition of a positive semi-definite matrix into two positive semi-definite matrices. ✖
- ☐ b. Locally Linear Embedding (LLE) is a linear embedding method that aims to preserve the neighborhood of each data point.
- ☒ c. Multidimensional Scaling (MDS) aims to find a low-dimensional representation of the data that preserves pairwise Euclidean distances. ✔
- ☐ d. Stochastic Neighbor Embedding (SNE) defines the similarity of two data points as a conditional probability while t-SNE defines it as a joint probability.
- ☒ e. Isomap measures distances between data points as the length of the shortest path along a neighborhood graph. ✔

C, D, E correct

Tick the correct statements about scaling and projection methods (several may be correct). We adapt to the notation from the slides and the lecture, i.e. we collect the original data points in a matrix X , and the mapped (e.g. down-projected) points in a matrix Y .

- ☒ The t-distribution has heavier tails than the Gaussian distribution, thus in t-SNE the joint distributions q_{ij} between y_i and y_j are computed using a t-distribution in order to alleviate the crowding problem.
- ☒ Isomap tries to approximate the geodesic distance between two points on a manifold by constructing a neighborhood graph.
- ☐ In SNE and t-SNE, the perplexity term is computed using the entropy of the similarities $q_{j|i}$ of the mapped data points y_i and y_j .
- ☐ In SNE and t-SNE, the objective is the mean squared distance between x_i and y_i .
- ☐ Locally linear embedding and multidimensional scaling both are probabilistic methods that rely on minimizing the KL-divergence between the distributions of X and Y .
- ☒ In SNE and t-SNE, the computation of the similarity $p_{j|i}$ between x_i and x_j relies on assuming a Gauss distribution centered at x_i .

A,B, F correct

- In t-SNE, the perplexity term is related to the effective number of neighbors and is computed from the conditional probabilities in the high-dimensional space, not directly from the similarities in the low-dimensional space.
- The objective in SNE and t-SNE is to minimize the Kullback-Leibler (KL) divergence between the probability distributions in the high-dimensional space and the low-dimensional space, not the mean squared distance.
- LLE and MDS are not probabilistic methods and do not rely on minimizing KL-divergence. LLE preserves local linear relationships, while MDS aims to preserve pairwise distances.

Non-negative Matrix Factorization (NMF)

Which statements about Non-negative Matrix Factorization (NMF) are correct?

As in the lecture, we have:

$$X = Y U^T$$

$$Y \in \mathbb{R}^{n \times l}$$

$$U \in \mathbb{R}^{m \times l}$$

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. NMF objectives are the KL divergence and the Euclidean distance. ✗
- ☐ b. NMF is always exactly solvable. ✗
- ☒ c. $X \in \mathbb{R}^{m \times n}$. ✓
- ☒ d. The non-negativity constraints make the representation of the observations purely additive. ✓
- ☒ e. The outer product representation reads $X = \sum_{k=1}^l y_k u_k^T$. ✓
- ☐ f. NMF factorizes any positive or negative definite matrix into two positive definite matrices. ✗

A,D, E is correct

$$X \in \mathbb{R}^{n \times m} \quad Y \in \mathbb{R}^{n \times l} \quad U \in \mathbb{R}^{m \times l}$$

$$0 \leq X_{ij} \quad 0 \leq Y_{ik} = [y_k]_i \quad 0 \leq U_{jk} = [u_k]_j$$

$$X = Y U^T = \sum_{k=1}^l y_k u_k^T$$

Mixture Models:

Mixturetown has two bakers call Albert (A) and Bibi (B). Both bake bread where the weights of the loafs they produce are distributed according to two Gaussian distributions.

The weight distributions of Albert's and Bibi's loafs are $f(x; \mu_A, \sigma_A)$ and $f(x; \mu_B, \sigma_B)$ respectively, where $f(x; \mu, \sigma)$ is the probability density function of a normal distribution with mean μ and standard deviation σ .

The means and standard deviations are known to be $\mu_A = 1$ kg, $\sigma_A = 0.1$ kg, $\mu_B = 1.5$ kg, $\sigma_B = 0.2$ kg.

Thus the distribution of loaf weights in Mixturetown is well described by a Mixture of two Gaussians:

$$p(x) = \alpha_A f(x; \mu_A, \sigma_A) + \alpha_B f(x; \mu_B, \sigma_B)$$

75 per cent of the loafs in Mixturetown are made by Bibi.

Somebody is so kind to give you a loaf of bread that weighs 1.1 kg.

Determine the probability that Bibi made this loaf of bread (up to two decimal places).

Some relevant quantities are $f(1.1; 1, 0.1) = 2.41971$, $f(1.1; 1.5, 0.2) = 0.26995$.

Answer:



The correct answer is: 0.25076529093399

To determine the probability that Bibi made the loaf of bread weighing 1.1 kg, given the Gaussian distributions of the weights from Albert and Bibi, we can use Bayes' theorem.

Given Data:

- Mean and standard deviation of Albert's loaves: $\mu_A = 1$ kg, $\sigma_A = 0.1$ kg
- Mean and standard deviation of Bibi's loaves: $\mu_B = 1.5$ kg, $\sigma_B = 0.2$ kg
- Probability that a loaf is made by Bibi: $P(B) = 0.75$
- Probability that a loaf is made by Albert: $P(A) = 0.25$
- Mixture distribution: $p(x) = \alpha f(x; \mu_A, \sigma_A) + \beta f(x; \mu_B, \sigma_B)$ where $\alpha = 0.25$ and $\beta = 0.75$

Gaussian Density Functions:

- $f(x; \mu_A, \sigma_A) = 2.41971$ for $x = 1.1$ kg
- $f(x; \mu_B, \sigma_B) = 0.26995$ for $x = 1.1$ kg

Gaussian Density Functions:

- $f(x; \mu_A, \sigma_A) = 2.41971$ for $x = 1.1$ kg
- $f(x; \mu_B, \sigma_B) = 0.26995$ for $x = 1.1$ kg

Bayes' Theorem:

To find $P(B | x)$, we use:

$$P(B | x) = \frac{P(x|B) \cdot P(B)}{P(x)}$$

Where:

- $P(x | B) = f(x; \mu_B, \sigma_B)$
- $P(x | A) = f(x; \mu_A, \sigma_A)$
- $P(x) = \alpha f(x; \mu_A, \sigma_A) + \beta f(x; \mu_B, \sigma_B)$

Calculation:

1. Calculate $P(x)$:

$$P(x) = \alpha f(x; \mu_A, \sigma_A) + \beta f(x; \mu_B, \sigma_B)$$

$$P(x) = 0.25 \cdot 2.41971 + 0.75 \cdot 0.26995$$

$$P(x) = 0.6049275 + 0.2024625$$

$$P(x) = 0.80739$$

2. Calculate $P(B | x)$:

$$P(B | x) = \frac{P(x|B) \cdot P(B)}{P(x)}$$

$$P(B | x) = \frac{0.26995 \cdot 0.75}{0.80739}$$

$$P(B | x) = \frac{0.2024625}{0.80739}$$

$$P(B | x) = 0.2507$$

So, the probability that Bibi made the loaf of bread weighing 1.1 kg is approximately 0.25 or 25%.

Mixture model:

Assume the following case:

The prior probability of component j is 0.1.

The probability for observing a data sample $\mathbf{x}$ given the total parameter set θ is 0.2.

The likelihood for observing a data sample $\mathbf{x}$, given component j and corresponding parameter vector, is 0.5.

Compute the posterior probability of component j .

To compute the posterior probability of component j in a mixture model, we can use Bayes' theorem. The formula for the posterior probability $P(j|\mathbf{x})$ is given by:

$$P(j|\mathbf{x}) = \frac{P(\mathbf{x}|j)P(j)}{P(\mathbf{x})}$$

Where:

- $P(j)$ is the prior probability of component j .
- $P(\mathbf{x}|j)$ is the likelihood of observing data sample $\mathbf{x}$ given component j .
- $P(\mathbf{x})$ is the marginal likelihood of the data, which can be found by summing over all possible components:

$$P(\mathbf{x}) = \sum_k P(\mathbf{x}|k)P(k)$$

In this context, $P(\mathbf{x}|\theta)$ is the marginal likelihood $P(\mathbf{x})$. It represents the probability of observing the data sample $\mathbf{x}$ across all components of the mixture model.

In this specific case, we have the following values:

- The prior probability $P(j) = 0.1$.
- The likelihood $P(\mathbf{x}|j) = 0.5$.
- The marginal probability $P(\mathbf{x}|\theta) = 0.2$.

Since we are only given information about one component and its likelihood, we can assume that the marginal likelihood $P(\mathbf{x})$ has been calculated taking all components into account.

Using Bayes' theorem, we can substitute the given values:

$$P(j|\mathbf{x}) = \frac{P(\mathbf{x}|j)P(j)}{P(\mathbf{x})}$$
$$P(j|\mathbf{x}) = \frac{0.5 \times 0.1}{0.2}$$

Now, let's calculate the posterior probability:

$$P(j|\mathbf{x}) = \frac{0.05}{0.2} = 0.25$$

The data generation in a mixture of Gaussians model is the following:

First one of the Gaussians is chosen according to its weight in the mixture model. Then a sample from the according Gaussian is drawn.

You have fitted a Mixture of two Gaussians to 1D data which resulted in the model

$$p(x) = \alpha_1 f(x; \mu_1, \sigma_1) + \alpha_2 f(x; \mu_2, \sigma_2)$$

where $f(x; \mu, \sigma)$ is the density of a normal distribution with mean μ and standard deviation σ .

To assign a sample to a cluster (one of the Gaussians) we need to calculate the probability that it was "produced" by it. The obtained parameters are:

$$\alpha_1 = 0.3, \alpha_2 = 0.7, \mu_1 = 0, \sigma_1 = 1, \mu_2 = 3, \sigma_2 = 1$$

Some relevant densities are

$$f(1; 0, 1) = 0.242, f(1; 3, 1) = 0.054$$

Calculate the probability that the datapoint $x = 1$ has been produced by the first Gaussian (up to two decimal places).

Antwort:



Die richtige Antwort ist: 0,657609

The probability density function for the mixture model is:

$$p(x) = \alpha_1 f(x; \mu_1, \sigma_1) + \alpha_2 f(x; \mu_2, \sigma_2)$$

First, we calculate $p(x = 1)$:

$$p(1) = \alpha_1 f(1; \mu_1, \sigma_1) + \alpha_2 f(1; \mu_2, \sigma_2)$$

$$p(1) = 0.3 \times 0.242 + 0.7 \times 0.054$$

$$p(1) = 0.0726 + 0.0378$$

$$p(1) = 0.1104$$

Next, we calculate the posterior probability that the data point $x = 1$ has been produced by the first Gaussian using Bayes' theorem:

$$P(G_1 | x = 1) = \frac{\alpha_1 f(1; \mu_1, \sigma_1)}{p(1)}$$

$$P(G_1 | x = 1) = \frac{0.3 \times 0.242}{0.1104}$$

$$P(G_1 | x = 1) = \frac{0.0726}{0.1104}$$

$$P(G_1 | x = 1) \approx 0.657$$

To two decimal places, the probability that the data point $x = 1$ has been produced by the first Gaussian is approximately 0.66.



KL Divergence

Assume you have two random variables Y and Z , which are uniformly distributed on $[0, \theta_Y]$ and $[0, \theta_Z]$, respectively, i.e.

$$p_Y(x) = \begin{cases} \frac{1}{\theta_Y} & \text{if } x \in [0, \theta_Y] \\ 0 & \text{else.} \end{cases}$$
$$p_Z(x) = \begin{cases} \frac{1}{\theta_Z} & \text{if } x \in [0, \theta_Z] \\ 0 & \text{else.} \end{cases}$$

For $\theta_Y = e^4$ and $\theta_Z = e^{10}$, compute the KL divergence

$$D_{KL}(p_Y||p_Z) = \int_{-\infty}^{\infty} p_Y(x) \log[p_Y(x)/p_Z(x)] dx.$$

As for "log", use the natural logarithm.

You can use a pocket calculator. Round to two decimal places. Number format example: 0.67

• $D_{KL}(p_Y||p_Z) =$  .

$$D_{KL}(p_Y||p_Z) = \int_{-\infty}^{\infty} p_Y(x) \log \left(\frac{p_Y(x)}{p_Z(x)} \right) dx$$

Given that $p_Y(x)$ and $p_Z(x)$ are uniform distributions on $[0, \theta_Y]$ and $[0, \theta_Z]$ respectively, we have:

$$p_Y(x) = \frac{1}{\theta_Y} \text{ for } x \in [0, \theta_Y]$$
$$p_Z(x) = \frac{1}{\theta_Z} \text{ for } x \in [0, \theta_Z]$$

The KL divergence simplifies to:

$$D_{KL}(p_Y||p_Z) = \int_0^{\theta_Y} \frac{1}{\theta_Y} \log \left(\frac{\frac{1}{\theta_Y}}{\frac{1}{\theta_Z}} \right) dx$$

$$D_{KL}(p_Y||p_Z) = \int_0^{\theta_Y} \frac{1}{\theta_Y} \log \left(\frac{\theta_Z}{\theta_Y} \right) dx$$

Since $\frac{\theta_Z}{\theta_Y}$ is a constant, the integral becomes:

$$D_{KL}(p_Y||p_Z) = \log \left(\frac{\theta_Z}{\theta_Y} \right) \int_0^{\theta_Y} \frac{1}{\theta_Y} dx$$

$$D_{KL}(p_Y||p_Z) = \log \left(\frac{\theta_Z}{\theta_Y} \right)$$

Now, substituting the given values $\theta_Y = e^4$ and $\theta_Z = e^{10}$:

$$D_{KL}(p_Y||p_Z) = \log \left(\frac{e^{10}}{e^4} \right)$$

$$D_{KL}(p_Y||p_Z) = \log(e^{10-4})$$

$$D_{KL}(p_Y||p_Z) = \log(e^6)$$

$$D_{KL}(p_Y||p_Z) = 6$$

So, the KL divergence $D_{KL}(p_Y||p_Z)$ is 6.00.

Clustering

Which statements about Clustering are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☒ a. In affinity propagation, the cluster centers are always data points.
- ☐ b. Clustering (or Cluster Analysis) is both a regression and a classification technique.
- ☐ c. In both k -means and fuzzy k -means, the membership distribution is always given by a binary random variable.
- ☒ d. In mixture models, an observation is assigned to the component with the largest prior distribution.
- ☐ e. Top-down clustering relies on building the minimal spanning tree and then in each step removes the largest of the remaining edges.
- ☐ f. In a mixture of Gaussians, the maximum likelihood and the maximum posterior always have the same value.
- ☐ g. The agglomerative hierarchical clustering algorithm repeatedly fuses the two maximally distant clusters.
- ☒ h. Similarity-based clustering does not require to represent objects via feature vectors.

A, H correct

- K-means; Membership distribution is a binary random variable.
- Fuzzy k-means: Membership distribution is typically represented by values between 0 and 1.
- Agglomerative hierarchical clustering typically fuses the two most similar (least distant) clusters at each step, not the maximally distant ones.
- Top-down clustering, also known as divisive hierarchical clustering, doesn't directly involve building a minimal spanning tree (MST).
- The maximum likelihood estimation (MLE) maximizes the likelihood function, while the maximum posterior (MAP) estimation maximizes the posterior distribution, which includes the prior. They may not yield the same value unless the prior is uniform.
- Agglomerative hierarchical clustering typically fuses the two most similar (least distant) clusters at each step, not the maximally distant ones.

Which statements about Clustering are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. k -means is faster than MoG and not sensitive to initialization.
- ☐ b. Both k -means and MoG assign an observation to a single cluster only.
- ☐ c. The mixing coefficients in a Mixture of Gaussians model are represented by the posteriors.
- ☐ d. The maximum a posteriori estimation is a way to avoid near-zero variances and singular solutions in the maximum likelihood estimation of mixture models.
- ☐ e. Given an observation $\mathbf{x}$, a component j and a parameter vector θ , the posterior $p(j | \mathbf{x}, \theta)$ describes how likely it is that $\mathbf{x}$ was produced by j .
- ☒ f. The EM algorithm for MoG tries to maximize the likelihood with respect to the cluster centers, the covariance matrices and the mixing coefficients. ✓
- ☒ g. The k -means algorithm is a simplification of the Mixture of Gaussians model. ✓
- ☒ h. Clustering is an unsupervised technique that aims at finding separate regions of high data density called clusters. ✓

D,E,F,G,H are correct

Frage 6

Falsch

Erreichte Punkte 0,00 von 7,00

Frage markieren

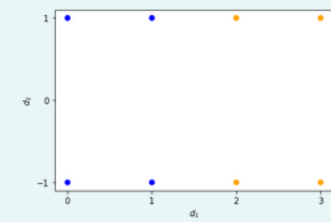
You are given 8 datapoints in $\mathbb{R}^2$. $\{(0, 1), (1, 1), (2, 1), (3, 1), (0, -1), (1, -1), (2, -1), (3, -1)\}$.

Agglomerative hierarchical clustering analysis was done on this data until only two clusters remained with different linkage functions, namely

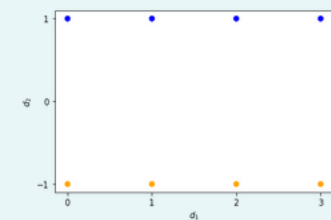
- single linkage: $d_{\min}(A, B) = \min_{a \in A, b \in B} \|a - b\|$
- complete linkage: $d_{\max}(A, B) = \max_{a \in A, b \in B} \|a - b\|$

One of the clusterings below has been generated using single linkage and one using complete linkage.

Clustering 1:



Clustering 2:



Check the correct statements:

- ☒ a. Clustering 1 has been obtained by single linkage ✖
- ☐ b. Clustering 2 has been obtained by single linkage
- ☒ c. Clustering 2 has been obtained using complete linkage ✖
- ☐ d. Clustering 1 has been obtained using complete linkage
- ☒ e. One of the clusterings could have theoretically been the result when using either of the two linkage functions. ✔

Your answer is incorrect.

Die richtigen Antworten sind:

Clustering 1 has been obtained using complete linkage.

Clustering 2 has been obtained by single linkage.

One of the clusterings could have theoretically been the result when using either of the two linkage functions.

B, D, E are correct

- Clustering 1 has been obtained using **complete** linkage, as it results in more **compact** clusters.
- Clustering 2 has been obtained using **single** linkage, which can result in more **elongated**, less compact clusters.

Biclustering

Which statements about Biclustering are correct?

Multiple answers are possible!

Wählen Sie eine oder mehrere Antworten:

- ☐ a. A common biclustering method is the variance maximization method.
- ☒ b. FABIA is a generative model. ✔
- ☐ c. FABIA defines biclusters as inner products.
- ☒ d. In FABIA, the likelihood is analytically intractable and requires, e.g., the (variational) EM algorithm. ✔
- ☐ e. Row and column elements must belong to exactly one bicluster or no bicluster at all.
- ☒ f. Biclustering simultaneously clusters the rows and columns of a matrix. ✔

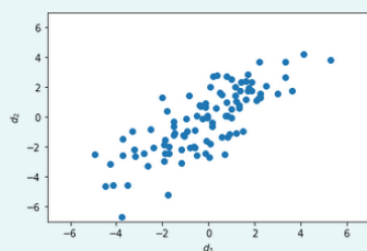
B,D, F correct

- FABIA uses a probabilistic model where biclusters are defined as the outer product of sparse factors.

Multivariate Gaussian

You are given n data points $\mathbf{x}_1, \dots, \mathbf{x}_n$ with $\mathbf{x}_i \in \mathbb{R}^2$. This data has been drawn from a multivariate Gaussian with density $p_{\text{true}} = \mathcal{N}(\mathbf{0}, \mathbf{C})$ with $\mathbf{C} = \begin{pmatrix} 5.5 & 4.5 \\ 4.5 & 5.5 \end{pmatrix}$.

The data are displayed in the figure below.



The likelihood of the data under the true density evaluates to $\mathcal{L}_{\text{true}}$.

You want to fit a factor analysis model with 1 factor to the data.

But instead of the usual factor analysis assumptions you drop the noise term.

That means your model assumption is that the data has been generated by first sampling $\mathbf{y}_i$ from a standard normal distribution, $\mathbf{y}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The $\mathbf{x}_i$ are generated by $\mathbf{x}_i = \mathbf{U}\mathbf{y}_i$, where $\mathbf{U} \in \mathbb{R}^{2 \times 1}$.

The maximum likelihood of the data for all possible values of $\mathbf{U}$ for this model is $\mathcal{L}_{\text{model}}$. Which of the following are correct:

- ☐ a. $0 \leq \mathcal{L}_{\text{model}} \leq 2\mathcal{L}_{\text{true}}$
- ☐ b. $\sqrt{n}\mathcal{L}_{\text{true}} \leq \mathcal{L}_{\text{model}} \leq n\mathcal{L}_{\text{true}}$
- ☒ c. $0 < \mathcal{L}_{\text{model}} < \mathcal{L}_{\text{true}}$ ✖
- ☐ d. $\frac{1}{n}\mathcal{L}_{\text{true}} \leq \mathcal{L}_{\text{model}} \leq n\mathcal{L}_{\text{true}}$

A is correct